

Jaemin Eom – Curriculum Vitae

Computational Robt Design & Fabrication (CREATE) Lab.
 Institute of Mechanical Engineering (IGM), School of Engineering
 EPFL (École Polytechnique Fédérale de Lausanne)
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Research Interests

- Design and Control of Robotic Grippers
- Computational Design for Robotic Hands
- Multi-Goal Motion Planning for Robotic Manipulation
- Soft Robotics
- Physics-Based Simulation
- Tendon-Driven Actuators
- Under-Actuated Systems

Experience

- Sep. 2025 - **Postdoctoral Research Associate**
 Present CREATE Lab, École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
 ♦ Advisor: Prof. Josie Hughes
- Mar. 2025 - **Postdoctoral Research Associate**
 Aug. 2025 Soft Robotics Research Center (SRRC), Biorobotics Lab, Seoul National University, Korea
 ♦ Advisor: Prof. Kyu-Jin Cho

Education

- Sep. 2017 - **Ph.D. in Mechanical Engineering** (GPA : 3.77/4.3)
 Feb. 2025 Seoul National University, Seoul, Korea
 ♦ Dissertation: Multi-Object Grasping Using Finger-to-Palm Translation for Pick-and-Place Tasks
 ♦ Advisor: Prof. Kyu-Jin Cho
 ♦ **Outstanding Doctoral Dissertation Award**
- Mar. 2013 - **B.S. in Mechanical Engineering** (GPA : 3.92/4.3)
 Aug. 2017 Seoul National University, Seoul, Korea
 ♦ **Summa Cum Laude**

PUBLICATIONS

International Journal

1. **Jaemin Eom**, Sung Yol Yu, Woongbae Kim, Chunghoon Park, Kristine Yoonseo Lee, and Kyu-Jin Cho, “MOGrip: Gripper for multiobject grasping in pick-and-place tasks using translational movements of fingers,” **Science Robotics (I.F. 26.1)**, vol. 9, eado3939, 2024. [\[Paper\]](#), [\[Video\]](#), [\[Project Page\]](#)
2. Yuna Yoo*, **Jaemin Eom***, MinJo Park, and Kyu-Jin Cho, “Compliant Suction Gripper with Seamless Deployment and Retraction for Robust Picking against Depth and Tilt Errors,” **IEEE Robotics and Automation Letters (I.F. 4.6)**, vol.8, no.3, 2023. **(Co-first author)** [\[Paper\]](#), [\[Video\]](#), [\[Project Page\]](#)
3. Woongbae Kim, **Jaemin Eom**, and Kyu-Jin Cho, “A Dual-Origami Design that Enables the Quasisequential

Deployment and Bending Motion of Soft Robots and Grippers,” **Advanced Intelligent Systems (I.F. 6.8)**, vol. 4, no. 3, 2021. [\[Paper\]](#), [\[Video\]](#)

4. Jun-Young Lee, **Jaemin Eom**, Sung Yol Yu, and Kyu-Jin Cho, “Customization Methodology for Conformable Grasping Posture of Soft Grippers by Stiffness Patterning,” **Frontiers in Robotics and AI**, vol. 7, 2020. [\[Paper\]](#)

Referred Conference Paper

5. Jun-Young Lee, **Jaemin Eom**, Woo-Young Choi and Kyu-Jin Cho, “Soft LEGO: Bottom-up Design Platform for Soft Robotics,” **2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**, 2018, pp. 7513-7520. [\[Paper\]](#), [\[Video\]](#)

Journals in Preparation

6. **Jaemin Eom**, and Kyu-Jin Cho, “Path Planning for Multi-Object Grasping in a Declutter Problem,” in preparation. [\[Project Page\]](#)
7. Haewoo Lee*, **Jaemin Eom***, JaeHyun Lee, and Kyu-Jin Cho, “Robotic Hand Design for Multi-Object Grasping of Column-Shaped Objects,” in preparation. **(Co-first author)** [\[Project Page\]](#)

Patents

8. Kyu-Jin Cho, **Jaemin Eom**, Yuna Yoo, MinJo Park, “Longitudinal Deployable Vacuum Suction Cup,” **PCT/KR2023/019352** (Patent Application, filed on Nov. 28th, 2023).
9. Kyu-Jin Cho, Jun-Young Lee, **Jaemin Eom**, “Soft Block Unit Comprising Expanding Block and Bending Block,” **JP Patent 6620257** issued Nov. 22th, 2019.
10. Kyu-Jin Cho, Jaemin Eom, Yuna Yoo, MinJo Park, “Longitudinal Deployable Vacuum Suction Cup,” **KR Patent 10-2624036** issued Jan. 8th, 2024.
11. Kyu-Jin Cho, **Jaemin Eom**, Sung Yol Yu, Woongbae Kim, “Gripper for Gripping Multi-Object with an Internal Storage,” **KR Patent 10-2497956** issued Feb. 6th, 2023.
12. Kyu-Jin Cho, Jun-Young Lee, **Jaemin Eom**, “Soft Block Unit Comprising Expanding Block and Bending Block,” **KR Patent 101950654** issued Apr. 14th, 2019.

Dissertation

13. **Jaemin Eom**, “Multi-Object Grasping Using Finger-to-Palm Translation for Pick-and-Place Tasks,” Seoul National University, Seoul, Korea. [\[pdf\]](#)

Honor and Awards

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| Apr. 2026 | Reviewer of the Year, <i>npj Robotics</i> (Nature Portfolio), 2025 (Official announcement expected May 2026) |
| Feb. 2025 | Outstanding Doctoral Dissertation Award (Department of Mechanical Engineering) |
| Apr. 2021 | 1 st prize winner, RoboSoft 2021 Manipulation Challenge, IEEE International Conference on Soft Robotics |
| Nov. 2020 | Silver Prize, 5 th KSME-SEMES Open Innovation Challenge, Young Engineers Group |
| Dec. 2019 | Silver Prize, 4 th KSME-SEMES Open Innovation Challenge, Young Engineers Group |
| Apr. 2019 | 3 rd prize winner, RoboSoft 2019 Manipulation Challenge, IEEE International Conference on Soft Robotics |
| Feb. 2019 | Bronze Prize, 25 th SAMSUNG Humantech Paper Award |
| Nov. 2023 | Honorable Mention, 8th KSME-SEMES Open Innovation Challenge (Young Engineers Group) |
| Nov. 2023 | Honorable Mention, 8th KSME-SEMES Open Innovation Challenge (Young Engineers Group) |

Professional Service

Peer Reviews (Total: 6)

- **Selected as “2025 Reviewer of the Year” by *npj Robotics***
- **Science Robotics: 1**
- **npj Robotics: 1**
- IEEE Robotics and Automation Letters: 2
- IEEE-RAS International Conference on Soft Robotics (RoboSoft): 1
- Intl. Conf. on Embodied Intelligence: 1

Research Experience

- Jan. 2026– Present **Computational Design and Structural Optimization of a Robotic Hand**
- Conducted optimal design of a robotic hand focused on maximizing grasping force and stability.
 - Utilized MuJoCo physics engine and Bayesian Optimization to determine the optimal kinematic parameters for robust manipulation.
- Sep. 2025– Present **Task Planning for Bimanual Manipulation Systems**
- Developed a task allocation algorithm to distribute manipulation workloads between dual-arm systems
 - Demonstrated the feasibility of the planned manipulation tasks through teleoperation
- Sep. 2024– Aug. 2025 **Manipulator path planning for multi-object grasping in declutter problem**
- Efficiently solved the decluttering problem by grasping and transporting multiple objects at once.
 - Proposed an algorithm to find the minimum path for decluttering all given objects.
- Mar. 2024– Aug. 2025 **Robotic hand design for multi-object grasping**
- Designed a robotic hand that sequentially grasps multiple objects, stores them in the palm, and transports them all at once.
 - Analyzed finger links' length and joint stiffness for target motion through analytic model and simulation.
 - Design the experiments and demonstrations.
- Aug. 2019– Dec. 2024 **Multi-object grasping using finger-to-palm translation for pick-and-place tasks**
- Proposed finger-to-palm translation as a key manipulation skill for multi-object grasping in pick-and-place tasks.
 - Presented a finger design enabling finger-to-palm translation.
 - Introduced a soft conveyor palm design capable of storing multiple objects simultaneously.
 - Designed the experiment, conducted experimental work, and performed demonstrations.
 - Conducted analytic modeling and ABAQUS simulation.
- Dec. 2020– Dec. 2022 **Compliant suction gripper with seamless deployment and retraction**
- Supervised a UROP student and submitted a paper to IEEE Robotics and Automation Letters.
 - Designed a deployable body of a suction cup.
 - Proposed a pneumatic circuit design for seamless deployment, picking, and retraction.
 - Designed the experiments and demonstrations.
- Jan. 2020 – Dec. 2022 **Development of a collaborative assistive robot arm utilizing foldable soft robot technology**
- Funded by *Ministry of Trade, Industry & Energy*
- Integrated the developed foldable gripper and the developed robotic arm.

- Nov. 2020 – **Dual-Origami Design that Enables the Quasisequential Deployment and Bending Motion**
 Dec. 2021 ♦ Designed the experiments and demonstrations.
- Jan. 2018 – **Development of modular gripper for small quantity production process**
 Dec. 2020 Funded by *Korea Institute of Machinery & Materials*
 ♦ Principal investigator of research project.
 ♦ Controlled the developed soft gripper using ROS communication.
 ♦ Developed a customized soft gripper with task specific designs.
- Sep. 2017 – **Development of fundamental soft robotics technology for advanced soft grippers**
 May 2020 Funded by *Ministry of Trade, Industry & Energy*
 ♦ Principal investigator of research project.
 ♦ Developed pneumatically actuated soft gripper for various objects, especially e-commerce.
 ♦ Controlled the developed soft gripper using ROS communication.
 ♦ Benchmarked the Amazon Picking Challenge to analyze feasibility of gripper.
- Jan. 2018 – **A hybrid gripper with pinching and suction grasp modes using a soft reconfigurable structure**
 Dec. 2019 ♦ Designed of a soft reconfigurable structure that functions as a gripper with fingers and transforms into a suction cup upon everting.
 ♦ Proposed a tendon routing for adaptive grasping.
- Jan. 2017 – **Soft LEGO: Bottom-up design platform for soft robotics**
 Dec. 2018 ♦ Proposed modular soft pneumatic actuator design compatible with LEGO.
 ♦ Utilized the ABAQUS simulation to predict the behavior of a soft pneumatic module.
 ♦ Designed the demonstrations.

Teaching Experience

- Sep. 2019 - Dec. 2019 **Teaching Assistant**
Dynamics (M2794.001200)
 Seoul National University
Supervisor : Prof. Kyu-Jin Cho
- Mar. 2018 - Jun. 2018 **Teaching Assistant**
Management in Mechanical Engineering 1 (M2794.004500)
 Seoul National University
Supervisor : Prof. Young-sang Yoo
- Feb. 2026 - Present **M.S. Thesis/Semester Project Tutoring** (Prof. Josie Hughes)
 Supervised **one** M.S. student for thesis research
 Mentored **one** M.S. student for semester project
- Jan. 2025 – Aug. 2026 **B.S Thesis/UROP Tutoring** (Prof. Kyu-Jin Cho)
 Feb. 2023 - Dec. 2024 Supervised the B.S. Thesis of **one** undergraduate student
 Dec. 2020 - Dec. 2022 Mentored **six** students for the Undergraduate Research Opportunities
 Sep. 2018 - Aug. 2019

Technical Skills

Design & Manufacturing, Simulation, Motion Planning, Embedded system, Control

1. Actuator System Design and Control (Tendon-driven systems for the under-actuated gripper, Pneumatic circuit design, Low-level control)
2. Motion Planning for Robotic Manipulator
3. Physics-Based Simulation and Analysis (ABAQUS, MuJoCo, MATLAB)
4. Various Prototyping Experiences (MOGrip, Robotic hand, Deployable suction gripper, Experimental setups)
5. CAD design (SOLIDWORKS)
6. Manufacturing (Elastomer molding for soft robot fabrication, Laser cutting, 3D printing, Heat press)
7. Circuit design (Eagle CAD)